HackSphere 3.0

GridLens

Microgrid demand forecasting and dispatch planning

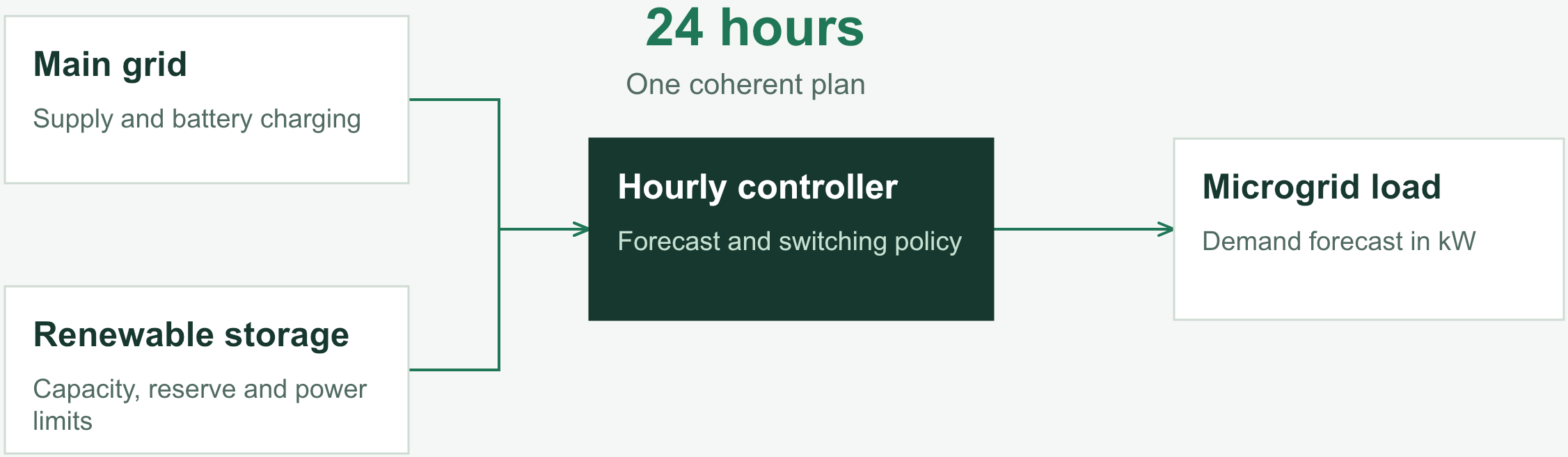


TEAM neevmodh205

Neev Modh
Solo participant

The microgrid planning challenge

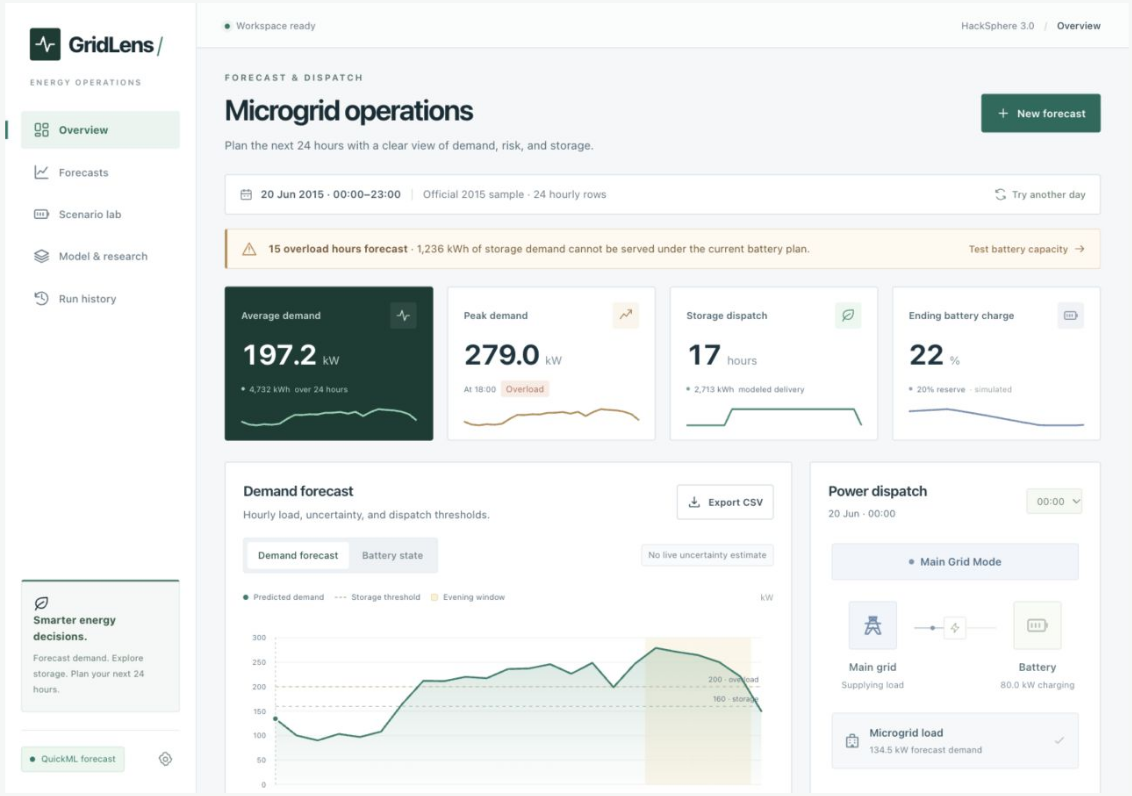
Weather changes demand. Storage has finite energy. Every hour needs an explicit decision.

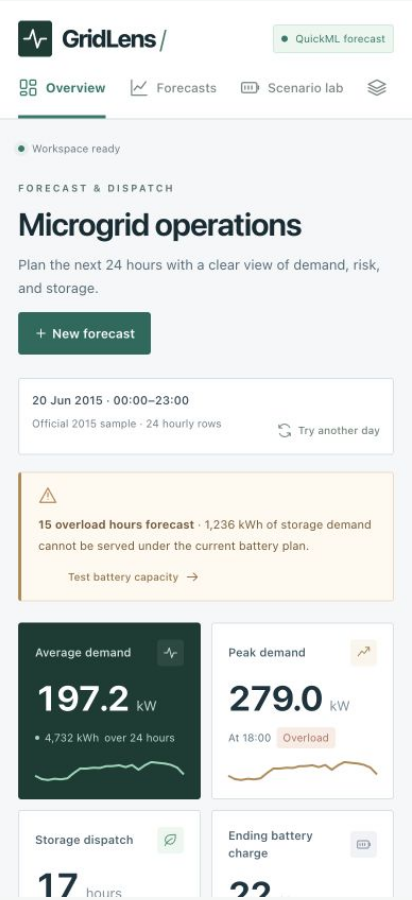


GridLens exposes overload risk and storage shortfall before a planned source transition.

A working operations dashboard

A complete forecast workflow on desktop and mobile



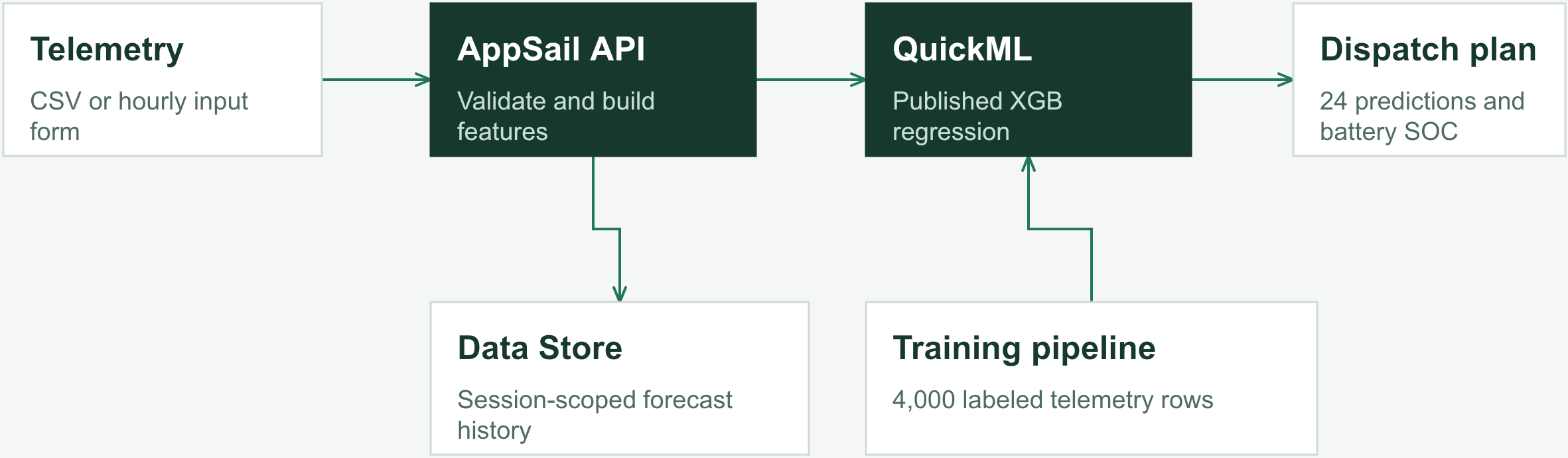


[Open the live dashboard]

Catalyst cloud architecture

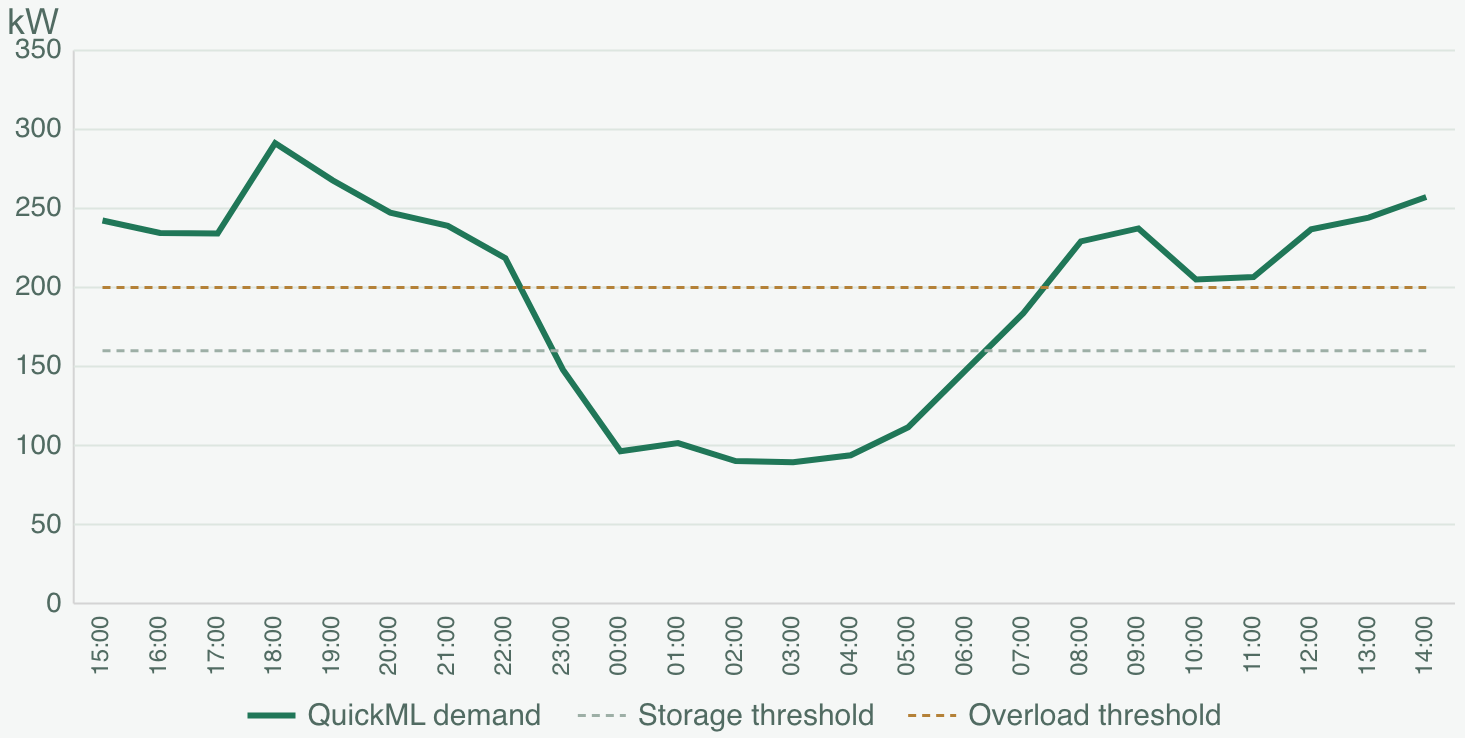
Shared feature preparation keeps training and inference consistent

Managed server credentials stay in Catalyst



Forecast and dispatch policy

Every hourly prediction produces a source decision and an independent overload check



STORAGE

Demand >160 kW
or 18:00–22:00 inclusive

MAIN GRID

All other hours

CRITICAL ALERT

Demand ≥200 kW
Independent of source mode

Real QuickML output: first 24 rows of the supplied sample input

Model performance and verification

Published QuickML XGB regression trained on 4,000 supplied labeled rows

8.27

RMSE / kW

6.54

MAE / kW

0.973

R^2

QuickML built-in evaluation. Hidden-test accuracy remains unknown.

1,000 rows

Original evaluator: SUCCESS
Live QuickML with local fallback disabled

36 tests

Rules, input cleanup, order preservation, session isolation and API behavior

Explore GridLens

Team neevmodh205

Neev Modh
Solo participant

Live application

github.com/neevmodh/GridLens

Forecasts and saved plans run on Catalyst.
Battery state and control commands are simulations.

